Just Keep Scrolling:

How Exposure to Short-Form Social Media Content Affects Recall in University Students

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Since the dawn of the internet in the late 1960s (Leiner et al., 2009), digital media has changed what it means to be human. It has restructured the way people work, the way they think, the way they study, and, perhaps more than anything else, the way they communicate. The first social media platform was released in 1997 (Pedrouzo & Krynski 2023), when digital communication had become more common and early cell phones had started to appear on the market. Six Degrees, aptly named after the mathematical theory of six degrees of separation, allowed users to connect with their friends, their friends' friends, and so on down the line, expanding the possible network of connections (Pedrouzo & Krynski 2023). The creation of Six Degrees inspired countless others: AOL, Facebook, Myspace, Reddit, and YouTube, to name a few. Since then, social media has become the de facto mode of communication for the younger generations: recent reports ("Global," 2024) have shown that daily social media use has gone up from 90 minutes in 2012 to 143 minutes in 2024.

Though digital communication has many benefits, connecting people across time zones and language barriers, it also has significant downsides. Shannon et al. (2022) found a correlation between excessive social media use and mental health issues such as depression and anxiety. Social media is harmful in the classroom, too. As demonstrated in several studies (Chossi et al., 2023; Khan et al., 2024; Spence et al., 2020), viewing short-form social media content while learning new information is highly detrimental to retention. Due to the damaging nature of social media and its prevalence among young adults, understanding its effect on students is crucial. This study seeks to examine whether viewing short-form, video-based social media content after learning new information has a negative impact on students' ability to recall that information. We will compare their responses to that of a control group, which will take a distraction-free break between being presented with the new information and being tested. We hypothesize that we will find a correlation between social media use and poorer performance on memory evaluations.

Effects of Social Media on Cognition and Mental Health

The negative side effects of social media use, both psychological and physical, have been studied across a variety of settings. Pedruozo and Krynski (2023) noted that these side effects can include physical lethargy, obesity, insomnia, cognitive issues, and worsened academic performance. A recent survey of Chinese high school students (n = 3036) focused on the effects of TikTok, reporting a mediating effect on memory loss between excessive TikTok use and psychiatric issues such as depression and anxiety (Sha & Dong, 2021). Similar results were observed in a study of Lebanese adults (Dagher et al., 2021), where social media use was strongly correlated with higher levels of anxiety. Social media use has also been proven to be associated with increased aggression (Khan et al., 2024). Though TikTok restricts its For You Page, which uses an algorithm to curate a hand-picked selection of short-form content, to users at least 16 years of age, minors commonly register as being 16 or older regardless of their real age. On the For You Page, they may be exposed to graphic and shocking content, as well as some of the many challenges that have spread across the internet, such as the sinister "Blue Whale Challenge" or the viral "Tide Pod Challenge," which encourage users to perform dangerous or reckless acts. Clearly, the psychological harm of social media for young adults cannot be underestimated. However, since there is no precedent to the internet age, many of the long-term dangers of social media exposure for children and adolescents remain unknown.

Memory Loss Associated With Social Media Use

Further studies have narrowed their focus to examine the effects of social media in academic settings. In their article "Short-Form Videos Degrade Our Capacity to Retain Intentions: Effect of Context Switching On Prospective Memory," Chiossi et al. (2023) predicted that use of social media would impair students' ability to recall and perform cognitive tasks. Participants (*n* = 60) were assigned to one of three groups — a TikTok group, a Twitter group, and a YouTube group — each of which completed a round of tasks, took a ten-minute break to view their app of choice, and then completed another round of tasks. A fourth group, the control, completed the first round of tasks, took a ten-minute break with no distractions, then continued on to the second round. Chiossi et al. found that the TikTok group performed significantly worse on the procedural memory task than the other three groups, with a dramatic uptick in the rate of incorrect answers (2023).

Spence et al. (2020) designed a similar experiment, hypothesizing that exposure to social media would lower retention rates in university students studying new course material. Participants (*n* = 45) were asked to listen to a short story before being tested on the details. One experimental group listened to the story while viewing Instagram, another did so after viewing Instagram, and a control group was exposed without any distractions. The groups were then evaluated with the Weschler Memory Scale IV. Out of the two experimental groups, both showed lower retention than the control group, in particular the group that viewed Instagram while listening to the new material (2020). The study confirmed that Instagram had a negative effect on memory and indicated that students should limit their use of social media in academic settings. A review of 33 studies focusing on social media interruptions in the workplace reinforced these findings, reporting that the effect of an interruption depends on when it occurs (Zahmat & Zhang, 2023). Additionally, Zahmat and Zhang's review showed that, compared to other participants, cognitive disruptions had a particularly negative effect on students (2023).

Current Study Design and Hypotheses

As previously stated, our study hopes to evaluate the effects of short-form social media content on retention rates in university students. Other studies, several of which are outlined above, have focused on TikTok and its negative psychological impact on its users, from increased rates of depression and anxiety to worsened academic performance and recall ability. However, due to the recent, though short-lived, TikTok ban and subsequent issues with accessibility, we have chosen to center our study on Instagram Reels. Participants in an experimental group will read a short story, then scroll through their feed of Instagram Reels for two minutes. A control group will read the same story, but take a two-minute break with no distractions. Both groups will then be asked to recall details about the story in a short survey composed of multiple choice questions. As an added data point, participants will be asked to report their screen time, as well as their daily Instagram use.

It has been definitively proven that viewing social media during or after exposure to new material harms students' retention. We expect to see these findings replicated in our study. We hypothesize that the experimental group will not be able to fully remember the short story, with the two minutes of exposure to Instagram Reels damaging their ability to form new memories. We predict that the control group, on the other hand, will be able to accurately recall details about the story. If our hypothesis is correct, our study will offer further proof that social media use is harmful in academic environments and should be restricted for better scholastic performance. Given the current research on this topic, one could reasonably claim that the harms of social media outweigh the benefits; we hope that our study will help to prove this claim.

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